LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 7: If-Then: Teaching Sprites to Decide

Module 3 | 45 minutes | Conditionals and sensing | Grades 3–5 · ages 8–10

Objective: Use if-then and if-then-else blocks with sensing conditions (touching, key pressed) inside loops so sprites react to what happens.

Success criteria

- I can use an if block with a sensing condition.
- I can explain why game rules go inside forever.
- I can use if-else to choose between two actions.

Key vocabulary

- **condition**: A yes/no (true/false) question.
- **if-then**: Runs blocks only when the condition is true.
- **else**: The blocks that run when the condition is false.
- **sensing**: Blocks that check what is happening, like touching or key pressed.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Simon Says “if”.
5 min	**Watch**	Lesson video; pause on “Keep checking!”.
8 min	**Build along**	Don’t-touch-the-lava game.
15 min	**Perform**	Gem detector, lava Code Lab, maze runner, why-forever question.
8 min	**Probe**	New maze, Maya’s bug, write two game rules.
4 min	**Wrap-up**	Exit ticket; one student explains the maze runner.

Unplugged warm-up

Simon Says “if” (5 min): “If you are wearing blue, then stand up. If you have a sister, then clap, else wave.” Students act only when the condition is true for them.

Teaching the idea

Conditionals let programs choose. “if <condition> then” runs its inside blocks only when the condition is true; “if … else” chooses between two branches. Conditions are boolean (hexagon) blocks, often from Sensing. Because an if checks once when it runs, game rules sit inside a forever loop.

Analogy: An if block is a guard at a gate: it asks a question, and only lets the blocks through when the answer is yes.

Live demo: Build the lava rule without a forever loop and play: it fails. Ask students why, then wrap it in forever. Then run the maze-runner animation and change the maze — same code works.

Questions to ask

Ask	Listen for
What is the condition here?	The yes/no question in the hexagon.
What happens when it is false?	The inside is skipped (or the else runs).
Why did the rule only work once?	The if wasn’t inside a forever loop.

Differentiation

Support
Give the lava rule as a printed script; students build it and change only the say text. Use the gem detector before the maze.

Stretch
Add a moving lava block (forever glide), or an if-else that makes the player invisible while space is pressed.

Common misconceptions

Misconception	How to fix it
An if keeps checking by itself	It checks once when it runs; use forever to keep checking.
The else part always runs	Only one branch runs: inside if true, else if false.
The condition is an action	Conditions ask questions; actions go inside the if.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

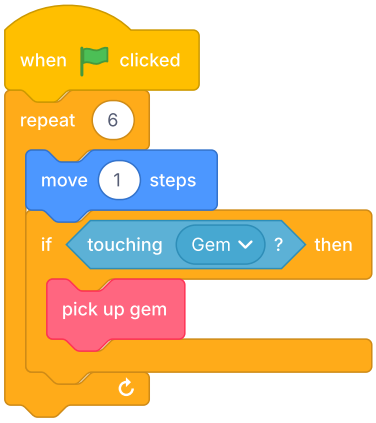
Exit ticket
Complete: “IF _____ THEN _____” for the lava game, and say where the if block must go.

Homework
Write three if-then rules for a board game you know.

Answer key — Kitabash activities

1. Gem detector (perform · block code)

One solution:

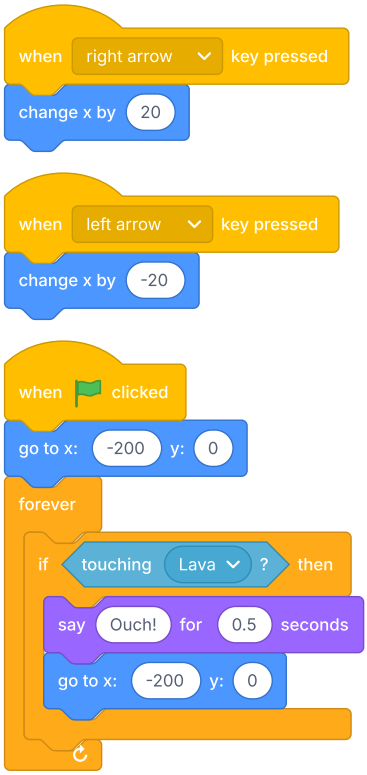


2. Code Lab: don't touch the lava (perform · code lab)

Goals checked: Uses an if block · Keeps checking (forever) · Says "Ouch!" on the lava · Goes back to the start

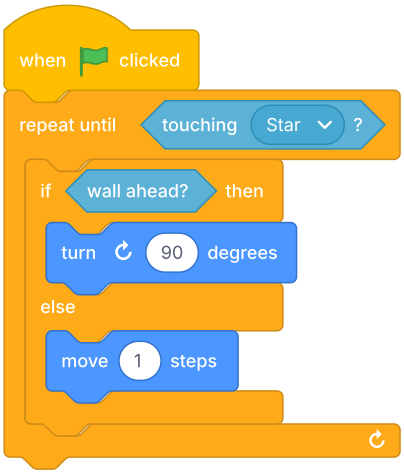
One solution:

Pip



3. Maze runner (perform · block code)

One solution:

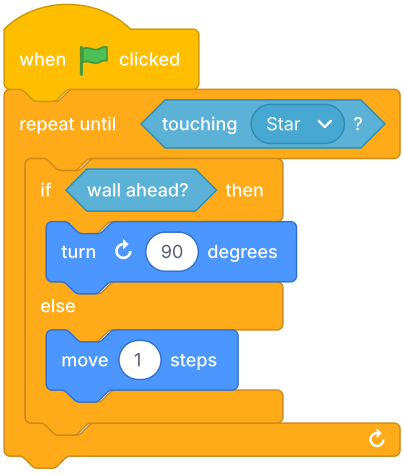


4. Why forever? (perform · multiple choice)

Answer: So the sprite keeps checking all through the game

5. New maze, same code (probe · block code)

One solution:



6. Debug it (probe · multiple choice)

Answer: Put the if inside a forever loop

7. Explain your thinking (probe · free response)

Look for: if, then, touch. **Example:** If the cat touches the mouse, then the score goes up by 1. If the cat touches the water, then it says “Yuck!” and goes back to the start.

Answer key — worksheet

1. **Real-life ifs:** Sensible actions, e.g. wear a jumper; go to class; eat, else keep playing.
2. **True or false?:** T · F
3. **What runs?:** It moves 5 steps (the else branch).
4. **Fix the rule:** Put the if block inside a forever loop.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.